

Optimization of Aggregates in Concrete Brick with Recycled Materials from Stone

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Abstract: Currently, conventional burnt clay brick makes up 95% of the total use of construction brick in Vietnam. There is an urgent need to use alternative materials for sustainable development. In this paper, we experimentally optimize the proportion and sizes of aggregates to produce concrete brick with recycled materials from stone mining. Experimental results showed that the optimum proportion of the aggregates of the concrete brick are: 50% chippings stones, 30% young stones, 8% cement, 12% fly ash and silica fumes; and the optimum sizes of the aggregates are 3.0 mm chippings stones and 2.5 mm young stones. The compressive strength of the concrete brick with the optimum mixture is 9.55 MPa, well above the requirement of the building code TCVN 6016-2011 of Vietnam. The use of local, cheap and abundantly available recycled materials significantly lowers the cost of the concrete brick. The results of this study were applied to the mass production of concrete brick by Asia 96 New Material Joint Stock Company in Vietnam.

Keywords: concrete brick, compressive strength, chipping stones, young stones, upland soil, fly ash, silica fumes

I. INTRODUCTION

According to a report from the Vietnamese government, burnt clay brick is currently the most used in Vietnam, about 95% of the total use of construction brick [1]. This is a big issue since the production of burnt clay brick produces a huge amount of CO₂. Much effort was made by the government to promote the use of other types of green alternatives. This research is stemmed from a national research funding to find the optimum proportion and sizes of aggregates to produce the concrete brick with recycled materials from stone mining process.

Many types of recycled aggregates were studied as replacements for traditional aggregates in concrete over the past few decades. Recycled aggregate concrete (RAC) has been shown to have a great impact on the environment and economy [2-3]. However, the use of RAC in the construction industry remains limited [4]. A number of studies have worked on the topics to establish the compressive strength and other mechanical properties with different replacements ratios of various recycled aggregates [5-12]. Sim et al. [13] showed that the particle size of coarse aggregate is an important parameter to the behavior of aggregate concretes. However, there are very few studies investigating the effect of particle sizes and proportion of coarse aggregates on the

behavior of concrete in the form of a brick.

This paper presents an experimental study with the aim to produce the concrete brick with recycled materials from stone mining with the compressive strength that satisfies the building code TCVN 6016-2011 of Vietnam. Experiments were conducted to optimize the sizes of the aggregates and their proportions.

This paper is organized as follows. In the next section, the properties of each individual aggregate in the concrete brick are investigated. Next, the procedure to make the brick is detailed. Next, we show the experiment to optimize the sizes and the proportion of the aggregates in the concrete brick. The final section is the conclusions.

II. MATERIALS

A. Young stones

Young stones (Fig. 1a) are formed from the stones that induced mechanical and chemical weathering. Agents such as ice, water, cracks or heat cause an expanding effect, widening the cracks in the stone and breaking the stones into pieces. The effect of mechanical weathering increases the exposed surface area of stone, which speed up the chemical weathering process. The original stones break into smaller pieces with lower compressive strength than the original stones.

B. Upland soil

Upland soil (Fig. 1b) is a type of terrain that is formed by weathering, denuding from mountains. The hills have small slopes and high organic residues due to the presence of plants. The height of hills is usually less than 200 meters. This type of soil has low economic value, therefore using it as raw material for brick production will lower the cost of the product. Once the upland soil reaches the standard of cleanliness, it is shoveled into the trough through the conveyor system. Upland soil materials used for the production of concrete brick must be carefully selected to be free of impurities.

C. Chippings stones

Chippings stones (Fig. 1c) used for the production of concrete bricks are waste products from stone mining. Based on the results of the experiments to determine impurity content in chippings stones, we found that the content of impurities is quite large, about 6.9%. It is needed to lower the number of impurities in the chippings stones to less than 3% in order to reach the strength requirement of the concrete



brick. The chemical composition of the chippings stones is shown in Table 1.

Table 1. Chemical composition of the chippings stones

	Al ₂ O ₃	SiO ₂	Fe ₂ O ₃	SO ₃	CaO	LOI	K ₂ O	Na ₂ O	MgO
%	18.26	60.12	4.02	1.10	3.48	1.30	4.86	2.44	2.05

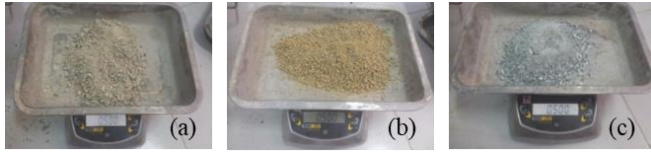


Figure 1. Image of (a) young stones, (b) upland soil and (c) chippings stones after grinding

D. Adhesives

PCB40 Portland cement is used as adhesive in this study. Portland cement is an inorganic binder in water, produced by calcining minced limestone (75% to 80%) and clay (20% to 25%) by heating block (about 1450°C), fast cooling to form clinker; then grinding fine clinker with plaster (3% to 5%) and some other essential additives such as inorganic additives. Table 2 describes the mineral composition of Portland cement.

Table 2. Mineral composition of Portland cement

Name of the mineral	Chemical formula	Abbreviation	%
Silicate tricalcite	3CaO.SiO ₂	C ₃ S	45 - 60
Silicate bicalcite	2CaO.SiO ₂	C ₂ S	20 - 30
Aluminate tricalcite	3CaO.Al ₂ O ₃	C ₃ A	4 - 12
Fero aluminate tetra calcite	4CaO.Al ₂ O ₃ .Fe ₂ O ₃	C ₄ AF	10 - 12
Aluminate calci	5CaO.Al ₂ O ₃	-	< 1
Fero aluminate calci	8CaO.3Al ₂ O ₃ .Fe ₂ O ₃	-	< 1
Ferit calci	2CaO.Fe ₂ O ₃	-	< 1

E. Fly ash

Fly ash is spherical and sharp, as shown in the scanning electron microscope (SEM) in Fig. 2. The mean particle size is about 3.74 μm , as described by the particle size distribution chart in Fig. 3 and Fig. 4. Adding fly ash to the mixture of the concrete brick helps to increase the strength while reducing the cost of the product. Thanks to its small size, the fly ash fills in the micro-holes of the cement particles, making the concrete brick denser, waterproof and higher in strength. The quantitative analysis of the composition of the fly ash is as follows: O (53.80), Na (1.84), Mg (1.06), Al (14.27), Si (20.25), P (0.09), S (0.19), K (0.25), Ca (0.94), Ti (1.95), Fe (4.72), Cu (0.16), As (0.15), Zr (0.34). The elements Si, Al and Ca have relatively high percentages, which makes the cement easy to hydrate, increasing the rate of cemented grains.

F. Silica fumes

Silica fumes are a type of binder that improves the mechanical properties of the mixture. They are mostly spherical, contained 95 wt.% of SiO₂, with an average diameter of about 1.0 μm , about one-hundredth the size of cement particles (Fig. 5). The chemical composition of silica fumes is shown in Table 3.

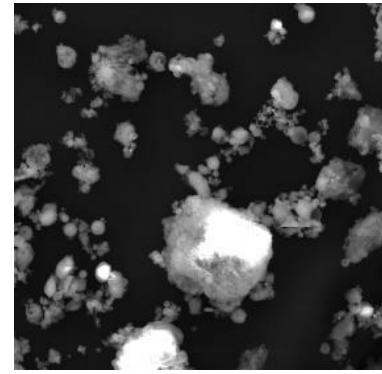


Figure 2. SEM image of fly ash

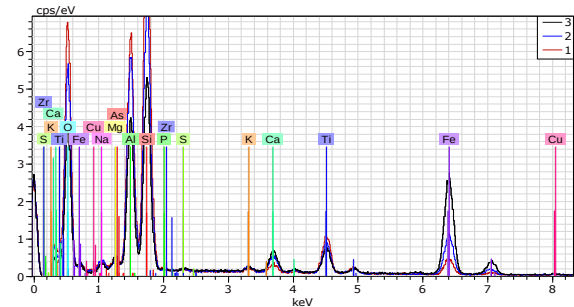


Figure 3. Spectral power distribution enlarged 2000 times

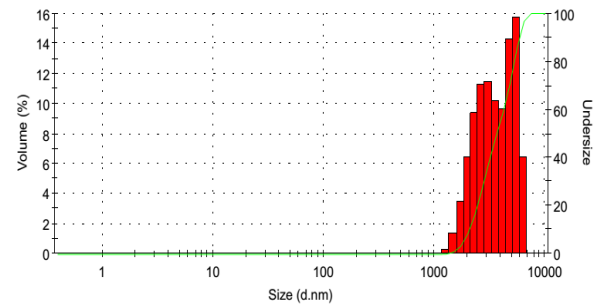


Figure 4. Graph particle size distribution of fly ash

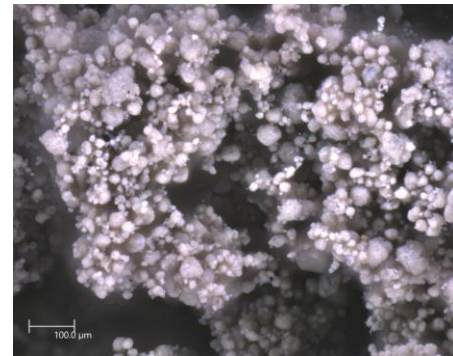


Figure 5. Image of silica fumes grains photographed using microscope

Table 3. Chemical composition of silica fumes

No	Oxide	%
1	SiO ₂	96.25
2	Al ₂ O ₃	0.23
3	Fe ₂ O ₃	0.07
4	CaO	0.31
5	MgO	0.04
6	K ₂ O	0.56
7	Na ₂ O	0.15
8	LOI	1.92

III. MANUFACTURING PROCESS

The process of manufacturing the samples with size 40 mm × 40 mm × 160 mm is described in Fig. 6.

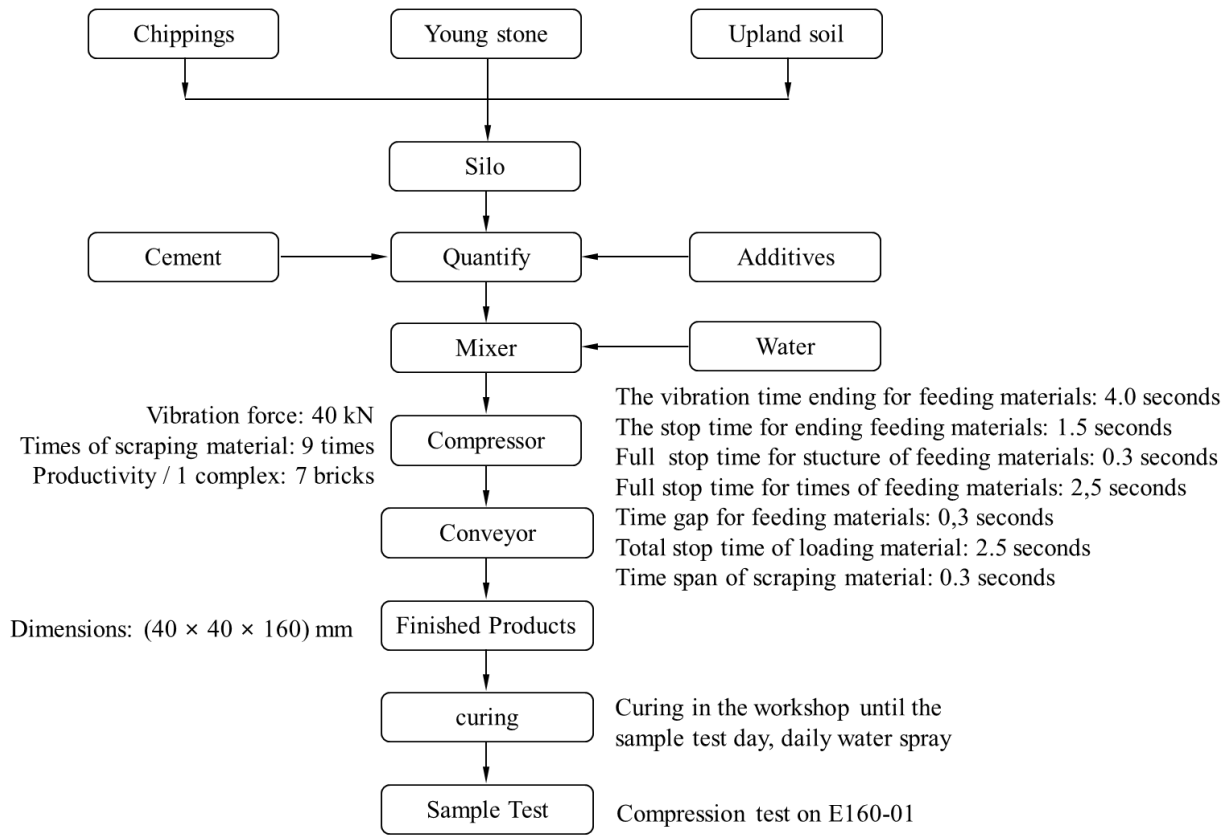


Figure 6. The manufacturing process of sample concrete brick

IV. EXPERIMENTAL STUDIES

Fig. 7a shows the mean compressive strength of the chippings stones with particle sizes from 1.0 mm to 5.0 mm using the fracture cracking test. For each particle's size, a total of nine samples were made and averaged. At the size of 3.0 mm, the mean compressive strength of the chippings stones is the highest (Fig. 7a). Similarly, it is found that at the size of 2.5 mm, the mean compressive strength of the young stones is the highest (Fig. 7b). Those optimum sizes of chippings stones and young stones are then used for following experiments to find the optimum proportion of aggregates of the concrete brick.

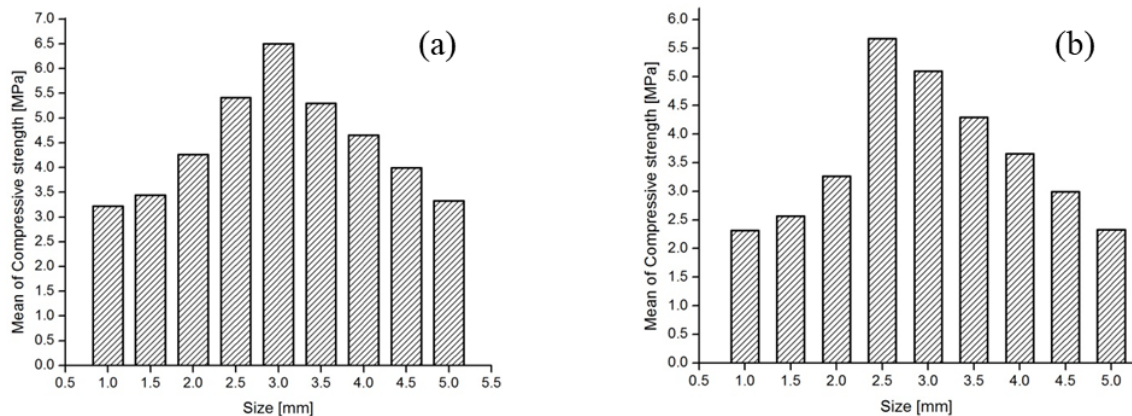


Figure 7. Mean values of compressive strength of (a) chippings stones and (b) young stones

Table 4. The mixing ratio and the compressive strength of the components

No	Chippings [%]	Young stone [%]	Upl-and soil [%]	Ce-me-nt [%]	Addi-tives [%]	Compressive strength [MPa]		
						7 days	14 days	28 days
M ₁	70	0	10	1	19	3.03	3.98	5.02
	70	0	10	3	17	3.32	3.94	4.98
	70	0	10	5	15	2.78	3.48	4.39
	70	0	10	8	12	2.63	3.33	4.21
M ₂	0	90	0	1	19	4.19	5.98	7.55
	0	90	0	3	17	4.83	6.01	7.59
	0	90	0	5	15	3.88	6.43	8.12
	0	90	0	8	12	4.82	5.72	7.22
M ₃	0	50	30	1	19	2.93	4.33	5.26
	0	50	30	3	17	3.89	4.81	6.27
	0	50	30	5	15	5.15	6.54	7.65
	0	50	30	8	12	4.76	5.75	6.82
M ₄	50	30	0	1	19	5.71	7.16	8.39
	50	30	0	3	17	6.32	7.57	9.02
	50	30	0	5	15	7.34	8.70	8.83
	50	30	0	8	12	6.91	8.33	9.55

V. CONCLUSION

In this paper, we optimize the proportion and sizes of aggregates to produce concrete brick with recycled materials from stone mining. Experimental results showed that the optimum size of the chippings stones and young stones are 3.0 mm and 2.5 mm, respectively. The optimum proportion of aggregates of the brick are 50% chippings stones, 30% young stones, 8% cement, 12% fly ash and silica fumes, which result in the strength of the concrete brick being 9.55 MPa, satisfying the building code TCVN 6016-2011 in Vietnam. The use of local, cheap and abundantly available recycled materials lowers significantly the cost of concrete brick. The results of this study were applied to the mass production of concrete brick by Asia 96 New Material Joint Stock Company in Vietnam. This study has the application not only in Vietnam but also in other places with similar recycled resources.

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